INCH-POUND

A-A-59697A
24 November 2014
SUPERSEDING
A-A-59697
1 October 2001

# COMMERCIAL ITEM DESCRIPTION

## COIL, RF, CHIP, FIXED, HIGH FREQUENCY

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers the general requirements for a radio frequency coil. Coils covered by this CID are intended for commercial/industrial applications.

2. CLASSIFICATION. This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see 7.1).

| AA59697 | - | 001 | F | T |
|---|---|---|---|---|
| CID number | | Dash number (see table I) | Inductance Tolerance (see table II) | (Optional) Termination finish (see table III) |

3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. Coils supplied to this CID shall be as specified herein. (see figure 1).

3.2. Electrical characteristics. The electrical characteristics shall be as specified in table I.

3.3 Weight. The weight shall be no greater than 0.5 gram maximum.

3.4 Operating temperature range. The operating temperature range is –40°C to +125°C.

3.5 Temperature rise. DC current rating for 15°C rise..

3.6 Altitude. The maximum altitude is 70,000 feet.

3.7 Marking. Coils supplied to this CID shall be marked with the manufacturer's standard commercial PIN.

3.8 Pure tin. The use of pure tin, as an underplate or final finish is prohibited both internally and externally. Tin content of transformer components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass.

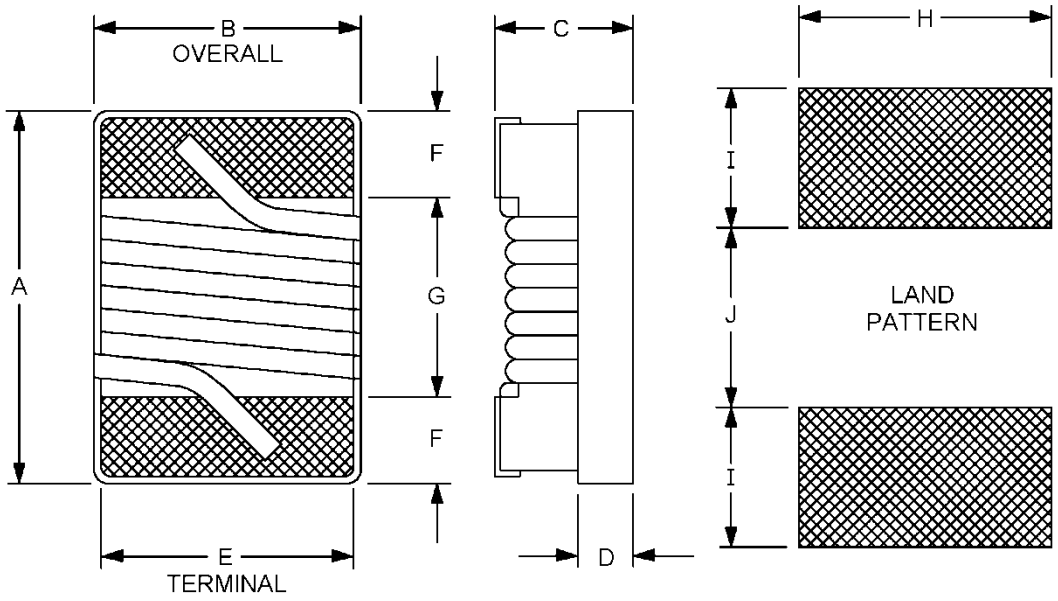
3.9 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.10 Workmanship. Coils shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: DLA Land and Maritime, Columbus, ATTN: VAT, P.O. Box 3990, Columbus OH 43218-3990, or email (transformers@dla.mil ). Since contact information can change you may want to verify the currency of the address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5950

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| Dimension | INCHES | MM |
|---|---|---|
| A | 0.115 Max. | 2.92 Max. |
| B | 0.110 Max. | 2.79 Max. |
| C | 0.080 Max. | 2.03 Max. |
| D | 0.020 Ref. | 0.51 Ref. |
| E | .0.080 | 2.03 |
| F | 0.020 | 0.51 |
| G | 0.060 | 1.52 |
| H | 0.100 | 2.54 |
| I | 0.040 | 1.02 |
| J | 0.050 | 1.27 |

NOTES:

1. Dimensions are in Inches.
2. Metric equivalents are listed for general information only.
3. Height Dimension “C” is before optional solder application. For maximum height dimension including solder, add 0.006 in (0.152 mm).
4. Terminal wrap around is approximately 0.010 (0.25 mm).

FIGURE 1. Configuration and dimensions.



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TABLE I. Electrical characteristics.

| CID dash number AA59697- | Inductance (nH) | Tolerance (±%) | Q min | SRF min (MHz) | DCR Max (Ohms) | $I_{DC}$Max (mA) |
|---|---|---|---|---|---|---|
| 001** | 10 @ 50 MHz | 5, 2 | 50 @ 500 MHz | 4100 | .08 | 1000 |
| 002** | 12 @ 50 MHz | 5, 2 | 50 @ 500 MHz | 3300 | .09 | 1000 |
| 003** | 15 @ 50 MHz | 5, 2 | 50 @ 500 MHz | 2500 | .10 | 1000 |
| 004** | 18 @ 50 MHz | 5, 2 | 50 @ 350 MHz | 2500 | .11 | 1000 |
| 005** | 22 @ 50 MHz | 5, 2, 1 | 55 @ 350 MHz | 2400 | .12 | 1000 |
| 006** | 27 @ 50 MHz | 5, 2 | 55 @ 350 MHz | 1600 | .13 | 1000 |
| 007** | 33 @ 50 MHz | 5, 2 | 60 @ 350 MHz | 1600 | .14 | 1000 |
| 008** | 39 @ 50 MHz | 5, 2 | 60 @ 350 MHz | 1500 | .15 | 1000 |
| 009** | 47 @ 50 MHz | 5, 2, 1 | 65 @ 350 MHz | 1500 | .16 | 1000 |
| 010** | 56 @ 50 MHz | 5, 2, 1 | 65 @ 350 MHz | 1300 | .18 | 1000 |
| 011** | 68 @ 50 MHz | 5, 2, 1 | 65 @ 350 MHz | 1300 | .20 | 1000 |
| 012** | 82 @ 50 MHz | 5, 2, 1 | 60 @ 350 MHz | 1000 | .22 | 1000 |
| 013** | 100 @ 25 MHz | 5, 2, 1 | 60 @ 350 MHz | 1000 | .56 | 650 |
| 014** | 120 @ 25 MHz | 5, 2, 1 | 60 @ 350 MHz | 950 | .63 | 650 |
| 015** | 150 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 850 | .70 | 580 |
| 016** | 180 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 750 | .77 | 620 |
| 017** | 220 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 700 | .84 | 500 |
| 018** | 270 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 600 | .91 | 500 |
| 019** | 330 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 570 | 1.05 | 450 |
| 020** | 390 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 500 | 1.12 | 470 |
| 021** | 470 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 450 | 1.19 | 470 |
| 022** | 560 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 415 | 1.33 | 400 |
| 023** | 620 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 375 | 1.40 | 300 |
| 024** | 680 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 375 | 1.47 | 400 |
| 025** | 750 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 360 | 1.54 | 360 |
| 026** | 820 @ 25 MHz | 5, 2, 1 | 45 @ 100 MHz | 350 | 1.61 | 400 |
| 027** | 910 @ 25 MHz | 5, 2, 1 | 35 @ 50 MHz | 320 | 1.68 | 380 |
| 028** | 1000 @ 25 MHz | 5, 2, 1 | 35 @ 50 MHz | 290 | 1.75 | 370 |
| 029** | 1200 @ 7.9 MHz | 5, 2 | 35 @ 50 MHz | 250 | 2.0 | 310 |
| 030** | 1500 @ 7.9 MHz | 5, 2 | 28 @ 50 MHz | 200 | 2.3 | 330 |
| 031** | 1800 @ 7.9 MHz | 5, 2 | 28 @ 50 MHz | 160 | 2.6 | 300 |
| 032** | 2200 @ 7.9 MHz | 5, 2 | 28 @ 50 MHz | 160 | 2.8 | 280 |
| 033** | 2700 @ 7.9 MHz | 5, 2 | 22 @ 25 MHz | 140 | 3.2 | 290 |
| 034** | 3300 @ 7.9 MHz | 5, 2 | 22 @ 25 MHz | 110 | 3.4 | 290 |



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TABLE I. Electrical characteristics. - continued

| CID dash number AA59697- | Inductance (nH) | Tolerance (±%) | Q min | SRF min (MHz) | DCR Max (Ohms) | $I_{DC}$Max (mA) |
|---|---|---|---|---|---|---|
| 035** | 3900 @ 7.9 MHz | 5, 2 | 20 @ 25 MHz | 100 | 3.6 | 260 |
| 036** | 4700 @ 7.9 MHz | 5, 2 | 20 @ 25 MHz | 90 | 4.0 | 260 |
| 037** | 5600 @ 7.9 MHz | 5 | 16 @ 7.9 MHz | 20 | 4.0 | 240 |
| 038** | 6800 @ 7.9 MHz | 5 | 18 @ 7.9 MHz | 40 | 4.9 | 200 |
| 039** | 8200 @ 7.9 MHz | 5 | 18 @ 7.9 MHz | 25 | 6.0 | 170 |

TABLE II. Inductance tolerances

| Code | Tolerance (percentage) |
|---|---|
| F | ±1 |
| G | ±2 |
| J | ±5 |

TABLE III. Termination finish.

| Code | Termination finish |
|---|---|
| Leave blank | Tin-lead |
| L | RoHS compliant, silver-palladium-platinum-glass-frit |
| T | RoHS, tin-silver-copper (95.5/4/0.5) |

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptance. The following market acceptance criteria are necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years.

b. The company must have sold 1,000 units meeting this CID in the commercial marketplace over the past 2 years.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format example.



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7.2 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmentally Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals is available on their website at http://www.epa.gov/epaoswer/hazwaste/minimize/chemlist.htm. Further information is available at the following EPA site: http://www.epa.gov/epaoswer/hazwaste/minimize/. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of the materials on the list should be minimized or eliminated unless needed to meet the requirements specified herein (see Section 3).

7.3 Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these inductors to DSCC under the Military Parts Control Advisory Group (MPCAG) evaluation program, CAGE code 58536 should be used.

7.4 Source of documents.

FEDERAL REGULATIONS

FAR - Federal Acquisition Regulations (FAR)

(Copies of these documents are available online at www.acquisition.gov/comp/far/index.html or from the U.S. Government Printing Office, 732 North Capital Street, NW, Washington D.C. 20401.)

7.5 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

7.6 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

c. Packaging requirements.

7.7 Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

| MFR's CAGE | MFR's name and address |
|---|---|
| 02113 | Coilcraft Inc<br>1102 Silver Lake Road<br>Cary, Illinois 60013-1658<br>Phone number (847) 639-2361<br>Uniform Resource Locator (URL): www.coilcraft.com |



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7.8 Part number (P/N) supersession data. This CID supersedes the following manufacturers' P/N's as shown. This information is being provided to assist in reducing proliferation in the government inventory system.

TABLE IV. P/N supersession data. 1/

| CID dash number (see table I) AA59697- | Vendor commercial PIN CAGE 02113 2/ | CID dash number (see table I) AA59697- | Vendor commercial PIN CAGE 02113 2/ |
|---|---|---|---|
| 001** | 1008CS-100X*L* | 021** | 1008CS-471X*L* |
| 002** | 1008CS-120X*L* | 022** | 1008CS-561X*L* |
| 003** | 1008CS-150X*L* | 023** | 1008CS-621X*L* |
| 004** | 1008CS-180X*L* | 024** | 1008CS-681X*L* |
| 005** | 1008CS-220X*L* | 025** | 1008CS-751X*L* |
| 006** | 1008CS-270X*L* | 026** | 1008CS-821X*L* |
| 007** | 1008CS-330X*L* | 027** | 1008CS-911X*L* |
| 008** | 1008CS-390X*L* | 028** | 1008CS-102X*L* |
| 009** | 1008CS-470X*L* | 029** | 1008CS-122X*L* |
| 010** | 1008CS-560X*L* | 030** | 1008CS-152X*L* |
| 011** | 1008CS-680X*L* | 031** | 1008CS-182X*L* |
| 012** | 1008CS-820X*L* | 032** | 1008CS-222X*L* |
| 013** | 1008CS-101X*L* | 033** | 1008CS-272X*L* |
| 014** | 1008CS-121X*L* | 034** | 1008CS-332X*L* |
| 015** | 1008CS-151X*L* | 035** | 1008CS-392X*L* |
| 016** | 1008CS-181X*L* | 036** | 1008CS-472X*L* |
| 017** | 1008CS-221X*L* | 037** | 1008CS-562XJL* |
| 018** | 1008CS-271X*L* | 038** | 1008CS-682XJL* |
| 019** | 1008CS-331X*L* | 039** | 1008CS-822XJL* |
| 020** | 1008CS-391X*L* | | |

1/ When "*" is used for coded values, it's the contractors responsibility to select those options allowed by the CID.

2/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID P/N shown.

7.9 Government users. To acquire information on obtaining these coils from the Government inventory system, contact DLA Land and Maritime, ATTN: FMTC, Post Office Box 3990, Columbus, OH 43218-3990, or telephone (614) 692-7727.

7.10 Changes from previous issue. The margins of this CID are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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MILITARY INTERESTS:

Custodians:
Navy - EC
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 7FXE

Preparing Activity:

DLA - CC

Project 5950-2013-012

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil .



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